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Triassic Paleomagnetism of Northern South America[1]

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Abstract Reconnaissance sampling of three radiometrically dated Late Triassic igneous units in northern South America provides new data on Triassic paleomagnetism and a new estimate of the position of the Triassic pole for South America. The new poles compare favorably with other Triassic poles from South America. Poles in Venezuela for Guacamayas Group (195 m.y.) and Bolivar dikes (199 m.y.), combined with the mean pole for the Las Cabras lavas (204 m.y.) of Argentina, give a mean pole for the South America Triassic at 72°S, 91°W, $\alpha_{95} = 15°$.

Bullard *et al.* showed that isobaths of the continental margins of Africa and South America have a minimum misfit for a rotation of 57.6° about a pole at 42.5°N, 29.9°W (500 fm isobath). Triassic paleomagnetic poles for Africa, when rotated about that same best-fit pole, merge with the Triassic poles for South America. A mean-composite pole for African and South American poles together is at 72°S, 106°W, $\alpha_{95} = 10°$, $N = 5$.

A dipole geomagnetic field is indicated over the South American and African landmasses during Triassic time.

Introduction

Paleomagnetic results are presented on Triassic rocks from the Guayana shield and El Baul uplift of Venezuela, and from the Santander massif of the Eastern Cordillera of Colombia. These specimens, collected during 1969–70, were intended to improve the definition of the Triassic magnetic field over the South American continent.[4]

Geologic Setting

Guacamayas Group—The El Baul uplift underlies an isolated hilly region rising above the plains (llanos) about 60 km south of the Caribbean Ranges and 130 km southeast of the Venezuelan Andes (Fig. 1). Deformed and lightly metamorphosed Cambro-Ordovician shales are intruded by the El Baul granite of Permian age (about 280 m.y.; Martin-Bellizzia, 1968). These Paleozoic rocks are overlain unconformably by acidic volcanic rocks, rhyolites, rhyodacites, and related volcaniclastic rocks of the Guacamayas Group (Martin-Bellizzia, 1961), for which a K-Ar whole-rock date of about 195 m.y. has been obtained (Table 1). This unit is approximately Late Triassic, according to the time-scale of Holmes (Harland *et al.*, 1964). The predominant deformation of these volcanic rocks is by faulting and tilting rather than folding (Martin-Bellizzia, personal commun., 1972). Although dips locally reach 90°, more moderate dips prevail.

Bolivar dike system—Several sets of diabase dikes are present in the northern part of the Guayana shield. We refer to the set which we sampled as the Bolivar dike system (Fig. 2). This narrow system of east-northeast-trending dikes extends from just north of Cerro Bolivar for more than 40 km eastward (Schwark *et al.*, 1971); its extent beyond this area is unknown. Cerro Bolivar is a prominent iron-mining center about 70 km south of Ciudad Bolivar and the Orinoco River. The Bolivar dikes intrude the Imataca gneiss, for which radiometric ages near 3,000 m.y. have been obtained. However, K-Ar whole-rock age measurements on samples from a site in these dikes indicate a Late Triassic

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[4] Symbols and abbreviations used in this report are: m.y., million years; oe, oersteds; a.f., alternating field; NRM, natural remanent magnetization; D, declination of paleomagnetic vector; I, inclination of paleomagnetic vector; p, colatitude of site relative to paleomagnetic pole; dp, radius of 95-percent confidence oval about paleomagnetic pole, in direction from pole toward site; dm, radius of 95-percent confidence oval in direction perpendicular to dp; $\alpha 95$, semiangle of cone of 95-percent confidence about a vector; κ, (kappa) Fisher (1953) precision parameter; and MDF, median destructive field.

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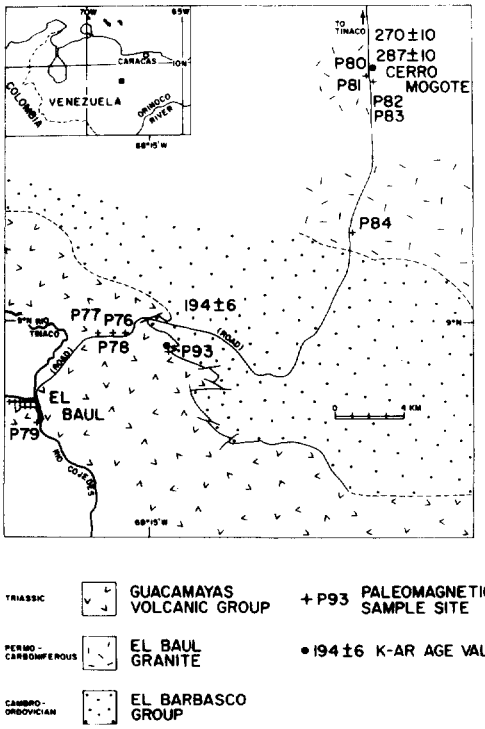


FIG. 1—Site-locality map, Guacamayas volcanic rocks, El Baul area, Venezuela.

age, near 198 m.y. (Table 1). Other dike systems, of unknown age, also are present in this region (McCandless, 1968). Especially abundant are dikes of northwest trend, northwest of Cerro Bolivar. The single age value obtained for the Bolivar dike system is assumed to be representative for the entire system of dikes extending along strike to the east-northeast. The close parallelism of the dikes and of paleomagnetic vectors for the dikes suggests a similar age for all sites sampled.

Pescadero rhyolite porphyry—Several granitic batholiths and associated plutons of Late Triassic to Middle Jurassic age are found in the resistant Santander massif east of Bucaramanga (Goldsmith *et al.*, 1971) near the juncture of the Cordillera Oriental of Colombia and the Merida Andes of Venezuela. Paleomagnetic results have been obtained for only one site, in the Pescadero rhyolite pluton, which has yielded a K-Ar date of 193 ± 6 m.y., near the Triassic-Jurassic boundary.

Sampling Procedures

Cores of 2.54-cm diameter were taken in the field with a portable gasoline-powered drill. Five or six cores were taken at each of four sites, and 19 cores were taken at a fifth site, in four flow units in the Guacayamas volcanic rocks. These sites are scattered over 15 km along the Tinaco-El Baul road (Fig. 1). From 4 to 10 cores were taken at each of five sites spaced over 40 km along the strike of the Bolivar dike system (Fig. 2). Access to these sites was by jeep and mining roads. Six cores were taken at the single site in the Pescadero rhyolite porphyry pluton, 11 km south of Los Aros, in outcrops in the Rio Manco, near 7°N, 73°W.

All cores were oriented by magnetic compass. No sun-compass measurements were possible in the Guacamayas volcanics or Pescadero rhyolite sites. For the Bolivar dikes, sun-compass readings were made on one to two cores at four of the five sites. The magnetic azimuths agreed with the sun-compass azimuth to within ± 2° at most localities. Magnetic azimuths were used for core orientations in all calculations.

TABLE 1. K-AR RADIOMETRIC DATES FOR VENEZUELAN SITES

Unit	Sec Ar^{rad}/gm ($\times 10^{-6}$)	Ar^{rad}/Ar^{tot}	%K	Age-date ($\times 10^6$y)
Guacamayas Group; rhyolite P93* 8.95°N, 68.28°W	13.22	0.74	1.64	192±3.8
	13.44	0.79	1.64	195±3.9
Bolivar dikes; diabase P99† 7.65°N, 63.43°W	5.71	0.37	0.694	199±5
	5.71	0.45	0.683	198±4

*At Finca El Socorro, near El Baul–San Carlos highway (see Fig. 1).

†In small quebrada adjacent to jeep road, locality Salsipuedes (see Fig. 2).

Analyst: Westwood Laboratories, Teledyne Isotopes, Westwood, New Jersey.

Constants: $\lambda_\beta = 4.72 \times 10^{-1}\ y^{-1}$; $\lambda_e = 0.585 \times 10^{-10}\ y^{-1}$;

$K^{40} = 1.19 \times 10^{-4}$ atom percent of natural potassium.

Correction: last column heading should be, ($\times 10^6$y).

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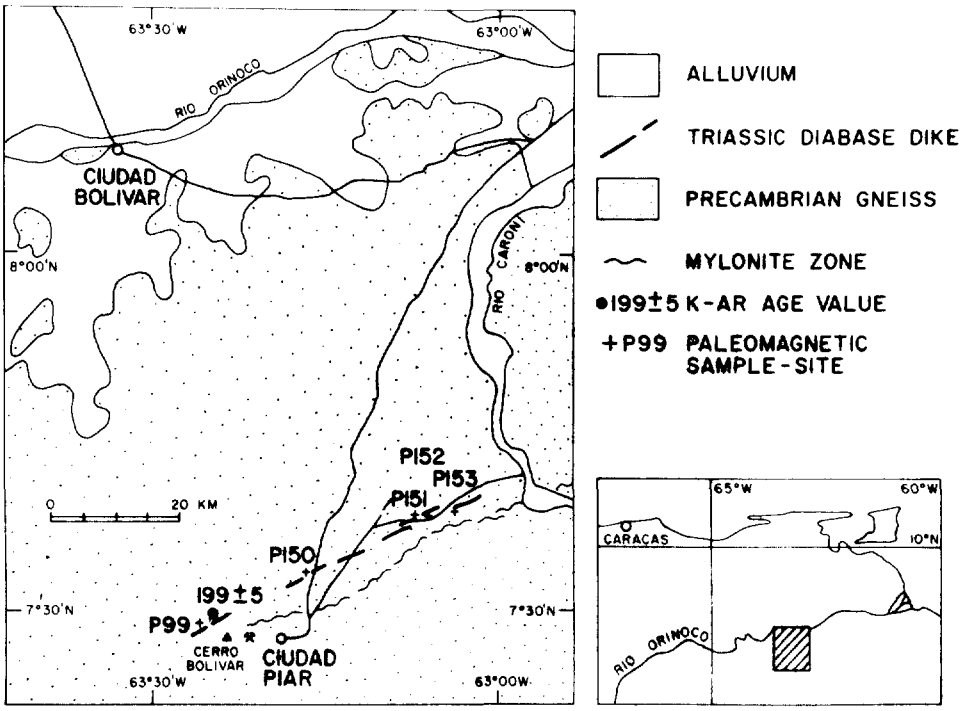


FIG. 2—Site-locality map, Bolivar dikes, Guayana shield, Venezuela.

Bedding measurements were possible and subsequent tilt corrections were necessary only for the Guacamayas volcanic rocks.

The cores, cut to a length of 2.54 cm, were measured on a commercially available 100-cps spinner magnetometer. Susceptibility measurements were made with a commercially available a.c. bridge. The magnetic properties of each of the rock units will be discussed in turn.

Natural-Remanent Magnetization and Alternating-Field Demagnetization

Bolivar Dikes

The NRM directions from these dikes are well grouped (Table 2) with declinations a little east of north with shallow positive or negative inclinations. The rocks are strongly magnetized, with intensities of magnetization of the order of 1×10^{-3} emu/cc and values of bulk susceptibility of about 2×10^{-4} emu/cc. Alternating-field demagnetization curves were plotted for one or more specimens from each site using demagnetization apparatus similar to that described by McElhinny (1966). Two of these demagnetization curves are shown in Figure 3. It can be seen from these curves that the intensity of magnetization does not fall off rapidly in low fields, the median destructive field being above 125 oe in one case and above 200 oe in the second. All sites except site 152 have MDF above 100 oe, which indicates good stability. The changes in direction are small during a.f. demagnetization, and the grouping of the directions is good as indicated by the high κ and small $\alpha 95$ (Table 2). Blanket demagnetization of all specimens was carried out in fields of 150 or 200 oe. The overall mean direction for the

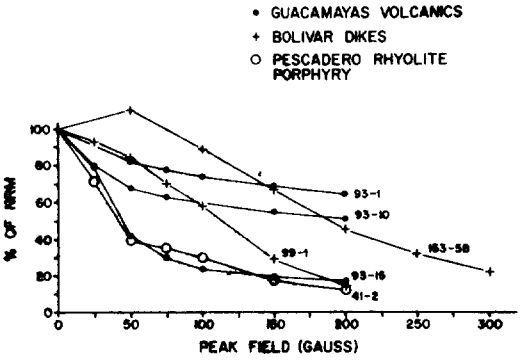


FIG. 3—A.f. demagnetization behavior of Triassic rocks from northern South America.

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TABLE 2. PALEOMAGNETIC SITE STATISTICS FOR TRIASSIC IGNEOUS UNITS, NORTHERN SOUTH AMERICA

Site	Lat. (N)	Long. (W)	NRM N	NRM D°	NRM I°	A.f. Demagnetized a.f. (oe)	A.f. Demagnetized D°	A.f. Demagnetized I°	A.f. Demagnetized κ	A.f. Demagnetized α95	Pole Position Lat. (S)	Pole Position Long. (W)	Pole Position dp	Pole Position dm
Bolivar dikes														
99	7°29'	63°18'	16	9.2	-0.3	150	14.9	-10.9	163.4	2.8°				
150	7°38'	63°17'	6	35.0	-22.3	150	22.0	-13.3	231.4	4.4°				
151	7°38'	63°07'	4	354.9	4.1	200	12.5	-18.5	138.2	7.8°				
152	7°32'	63°07'	5	35.3	-18.5	200	20.8	-16.4	203.2	5.3°				
153	7°29'	63°04'	10	21.0	7.5	150	19.4	-10.9	15.9	12.4°				
Average for unit	7.5°	63.2°	5				17.9	-14.0	245.6	4.9°	66.9°	114.4°	2.6°	5.0°
Guacamayas volcanics rocks*														
76	8°59'	68°15'	5	1.3	45.6	150	0.8	8.8	44.3	11.6°				
77	8°59'	68°15'	7	20.0	30.7	150		random						
78	8°59'	68°15'	7	102.9	61.6	150		random						
79	8°57'	68°15'	7	195.3	75.4	150		random						
93-I	8°57'	68°17'	7	351.9	26.8	150	349.3	-32.5	467.7	2.7°				
93-II	8°57'	68°17'	4	349.4	16.7	150	352.7	-30.4	288.5	5.4°				
93-III	8°57'	68°17'	5	358.0	18.8	150	351.2	-33.3	492.0	3.4°				
93-IV	8°57'	68°17'	3	5.2	42.1	150	8.5	-10.6	46.2	18.3°				
Average for unit	8.95°	68.25°	5				357.0	-19.9	16.8	19.2°	70.5°	59.3°	10.5°	20.1°
Pescadero porphyry														
41	6°49'	73°02'	9			200	359.6	17.6	22.9	10.9°	87.8°	83.2°	5.8°	11.2°

A.f. demagnetized results have been corrected for bedding rotation, but NRM results have not.

five sites after a.f. partial demagnetization is given in Table 2.

Pescadero Rhyolite Porphyry

In contrast to the Bolivar dikes, the single site from the Pescadero rhyolite does not show such stable behavior. The original direction of NRM is close to the present field direction toward the north with an inclination of 47°. Upon a.f. demagnetization the intensity of magnetization falls off rapidly, the MDF being 40 oe (Fig. 3). Above 50 oe, however, the drop in intensity is much more gradual indicating that a stable component is present in the rock. The principal change in direction occurs up to 50 oe and is reflected as a shallowing of the inclination. There is very little change in the direction of magnetization from 50 to 150 oe. Partial demagnetization of the entire site was carried out at fields of 100 and 200 oe, because the dispersion was least at 100 oe (Table 2).

Guacamayas Volcanic Rocks

Eight sites were sampled in individual flow units. In many of these sites the rocks had well-grouped directions of NRM with intensities near 4×10^{-3} emu/cc directed down and to the north. Several sites have widely dispersed directions of NRM. One or two specimens were chosen from each site and demagnetization curves were run on these specimens. Examples are given in Figure 3. These curves show that many of the sites exhibit extremely stable behavior. In the case of two of the samples figured the MDF exceeds 200 oe. On the basis of these curves, fields were chosen for overall demagnetization of the sites. After a.f. demagnetization the directions of magnetization at five of the sites are grouped very well with small circles of confidence. The site statistics are given in Table 2. After a.f. demagnetization, three of the sites are judged to be magnetized randomly, according to criteria of McElhinny (1966). The mean directions of magnetization for the Guacamayas volcanic rocks are given for bedding returned to horizontal, by rotations of 30° to 40° about the line of strike.

The directions of magnetization for the individual sites from all units, with their circles of 95-percent confidence, are shown in Figure 4. The statistical parameters are given in Table 2. The Guacamayas volcanic rocks and Bolivar dikes form two close but distinct groups, both of which have negative inclinations.

Mean Triassic Paleomagnetic Poles

Paleomagnetic poles for the three Triassic units of this study are given in Table 2. The mean of these three poles is 77°S, 89°W, $\alpha_{95} =$ 21°. A better estimate of the mean pole for

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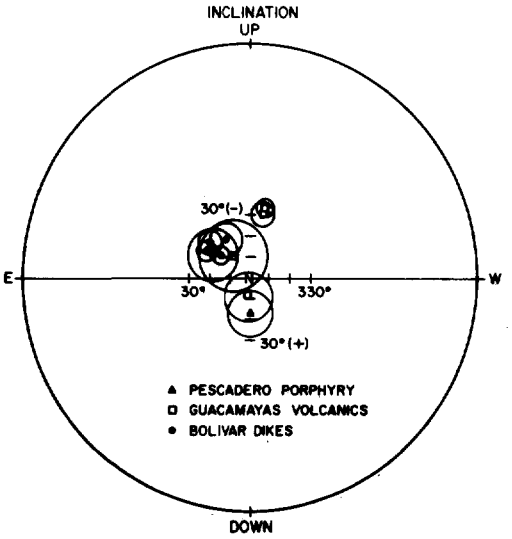


FIG. 4—A.f. demagnetization directions for Triassic rocks of northern South America. Equal-angle plot with prime representing vertical east-west plane, and local horizon straight line through center of projection. Demagnetized vectors of Guacamayas volcanic rocks and Bolivar dikes represent two close but distinct groups, with low, northerly directed, negative inclinations.

South America is obtained by incorporating all the magnetically stable and reliably dated units for South America. Thus, by including the pole for the Las Cabras lavas (204 m.y.) of Argentina (Valencio and Vilas, 1972) and the pole for dacitic tuff of the La Quinta formation (Late Triassic) of Venezuela (Hargraves and Shagam, 1969), a mean pole for five units of South America is obtained. That pole is at 74°S, 95°W, $\alpha_{95} = 10°$ (pole M2, Table 3). However, of these five poles, two might reasonably be excluded because of the paucity of sites represented. Omitting the La Quinta and Pescadero poles for that reason, the mean pole for the Triassic of South America is obtained from the remaining three units—Guacamayas volcanic rocks, Bolivar dikes, and Las Cabras lavas. That mean pole is at 72°S, 91°W, $\alpha_{95} = 15°$ (pole M3, Table 3 and Fig. 5A). The positions of all the poles involved in the preceding discussion are given in Table 4 and shown in Figure 5A.

Other poles of Triassic and possible Triassic age have been reported for South America. These are summarized in Table 4. They have been excluded from the calculation of the mean pole for South America for various reasons. Uncertainties in age, although in the Permian-Triassic range, can be attributed to the Giron Formation (Creer, 1970; Ward *et al.*, 1969; Langenheim, 1960), the Motuca Formation (Creer, 1970; Mendes and Petri, 1971), and the Paganzo redbeds and Salta-Jujuy beds (Creer, 1970). Similarly, the dolerite dikes of Suriname, dated at 227 m.y., fall on the Permian side of the Permo-Triassic boundary (Veldkamp *et al.*, 1971). Those dikes are approximately 20 m.y. older than the reliable Triassic units utilized for the calculation of the mean pole M3. Remagnetization of the Giron redbeds (Creer, 1970) and of the La Quinta redbeds (Creer, 1970; Hargraves and Shagam, 1967, 1969) also contributes uncertainty regarding the age of the magnetization of those units.

Comparison with Results from Africa

McElhinny *et al.* (1968) have examined Triassic paleomagnetic studies of Africa and have selected two poles, their B7 and B8, which satisfy their criteria of reliability. Subsequently, more recent reliable Mesozoic paleopoles have been published for Africa. Units from Morocco, studied by Hailwood and Mitchell (1971), are mainly in the Jurassic age range. Units from Sierra Leone (Briden *et al.*, 1971) may be Triassic, as they lie in the range 180–230 m.y. (late Early Jurassic to Late Permian). Because of the

TABLE 3. MEAN PALEOMAGNETIC POLES

Pole no.	Units	Paleomagnetic Pole			Reference
		Lat.	Long.	α95	
Mean pole for South America					
M1	1,2,3	77°S	89°W	21°	see Table 4
M2	1,2,3,4,5	74°S	95°W	10°	see Table 4
M3	1,2,4	72°S	91°W	15°	see Table 4
African poles					
B7	Shawa ijolite, Rhodesia (19.2° S, 31.7°E)	64°S	85.5°E	14°	McElhinny *et al.*, 1968; Gough and Brock, 1964
B8	Red sandstone Zambia (16.2°S, 28.7°E)	68°S	49.5°E	5°	McElhinny *et al.*, 1968; Opdyke, 1964
South America and Africa mean composite pole					
M4	1,2,4,B7*, B8	72°S	106°W	10°	Table 4 and this table
M5	1†,2†,3†, B7,B8	62°S	60°E	10°	Table 4 and this table

*Rotated −57.6° about best-fit pole of Bullard et al. (1965) at 42.5°N, 29.9°W; M4 is mean pole relative to present South America.

†Rotated 57.6° about best-fit pole at 42.5°N, 29.9°W; M5 is mean pole relative to present Africa.

Correction: Pole no. M4 should have B8*, not B8.

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TABLE 4. TRIASSIC AND POSSIBLE TRIASSIC PALEOMAGNETIC POLES OF SOUTH AMERICA

Pole	Unit	Site Location Lat.	Long.	N	Age ($\times 10^6$y)	Paleomagnetic Pole Lat.	Long.	p	dp	dm	Reference
Triassic											
1	Bolivar dikes, Venezuela	7.5°N	63.2°W	5	199	67°S	114°W	83°	2.6°	5.0°	this paper
2	Guacamayas Cp., Venezuela	8.95°N	68.25°W	5	195	71°S	59°W	78°	10.5°	20.1°	this paper
3	Pescadero rhyolite, Colombia	6.80°N	73.05°W	1	193	88°S	83°W	81°			this paper
4	Las Cabras lavas, Argentina	32°S	69°W		204	74°S	94°W		11°	18°	Valencio, 1969; Valencio and Vilas, 1972
5	La Quinta dacitic tuff, Venezuela	8°N	72°W	2	(Late Triassic)	69°S	109°W				Hargraves and Shagam, 1969; Creer, 1970
Triassic(?)											
6*	Motuca Formation, Brazil	6°S	43°W		(Permo-Triassic)	81°S	64°W				Creer, 1970
7†	Paganzo (III) red-beds, Argentina	29°S	68°W		(Permo-Triassic)	76°S	157°W				Creer, 1970
8†	Salta-Jujuy beds, Argentina	25°S	65°W		(Permo-Triassic)	67°S	105°W				Creer, 1970
9*	Giron Formation, Colombia	7°N	73°W		(Pennsylvanian-Jurassic)	77°S	106°W				Creer, 1970
10	Dolerite dikes, Suriname	4°N	55°W	10	227	82°S	40°W				Veldkamp et al., 1971

*Thermally demagnetized at 600°C.

†Natural remanent magnetization.

Correction: Pole no. 2, Guacamayas Gp.

unsuitable or uncertain age ranges of these units, however, their paleopoles have been excluded from calculations which are to follow on the Triassic mean paleopoles for Africa and South America.

Therefore, using the Triassic poles B7 and B8, the hypothesis will be tested that Africa and South America were part of a single landmass in Triassic time. A rotation which places the displaced continental margins in their best-fit position, prior to disruption, also should cause the synchronologic paleomagnetic poles of the two continents to be superimposed. Figure 5A shows the present distribution of the Triassic paleomagnetic poles of Africa and South America, in two distinct groups. Figure 5B shows that, when Africa is rotated into its best-fit position against South America, by a rotation of −57.6° about a pole at 42.5°N, 29.9°W (Bullard *et al.*, 1965), a similar rotation of the Triassic poles (B7 and B8) of Africa places them in a single group, together with the Triassic poles of South America. The mean of this single grouping, which can be regarded as the *mean composite Triassic pole* for the composite landmass Africa–South America, is 72°S, 106°W, $\alpha_{95} = 10°$, relative to the present geographic framework of South America. By rotating South America to Africa, the mean composite Triassic pole relative to the present geographic framework of Africa is 62°S, 60°E, $\alpha_{95} = 10°$. These mean composite poles, M4 and M5 respectively, are given in Table 3; M4 is shown in Figure 5B, together with the constituent poles of Africa and South America for which it is the mean.

Implicit in the calculation of the paleomagnetic-pole position for a site or for a unit is the assumption that the earth's paleogeomagnetic field was a dipolar field at the time of magnetization of the unit. This assumption seems to be valid for all five units of Africa and South America used to calculate the mean composite poles M4 and M5. We conclude that the geomagnetic field over South America and Africa was a dipolar field in Triassic time.

Bullard *et al.* (1965) calculated best-fit poles for different isobaths. The small number of paleomagnetic poles reviewed here is probably insufficient to evaluate their various poles and rotation angles for the net relative motion of Africa and South America. The minimum-misfit pole used here, representing the fit of the 500-fm isobaths, provides a good estimate of the net rotation (57.6° rotation about pole at 42.5°N, 29.9°W). Figure 5 indicates that the net rotation is post-Triassic.

Conclusions

New paleomagnetic data on radiometrically dated Late Triassic igneous rocks of South

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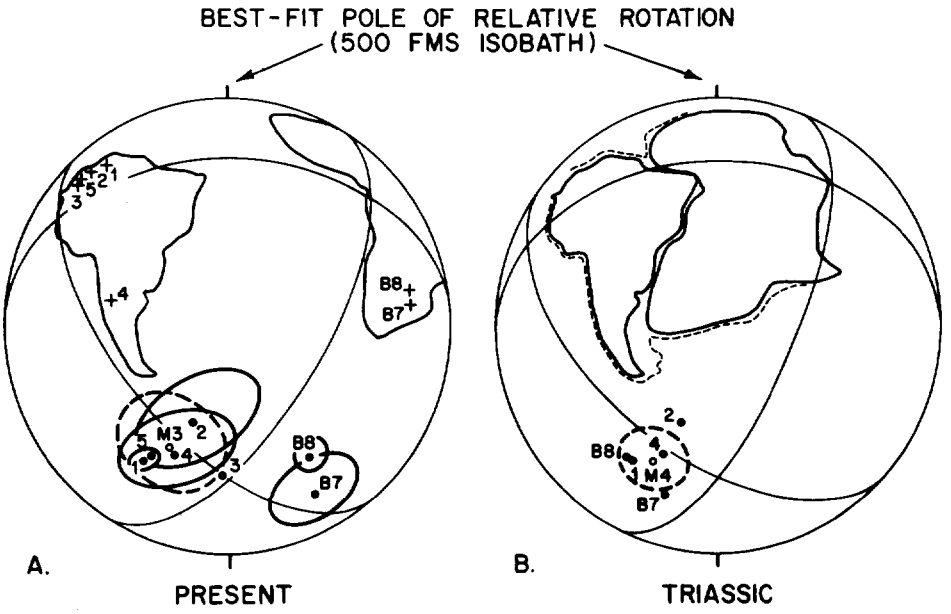


FIG. 5—Rotation about best-fit axis of rotation obtained by fitting continental margins also causes good fit for Triassic paleopoles of South America and Africa. *A*. Africa and South America, together with their corresponding Triassic paleopoles, shown in present relative positions. *B*. Africa and South America rotated to provide best fit of 500-fm isobaths; rotation of Africa is −57.6° about Euler pole at 42.5°N, 29.9°W (Bullard *et al.*, 1965). This rotation which provides best fit of continental-margin isobaths also draws Triassic poles of Africa into single group with those of South America. Orthographic projection, centered on 47.5°S, 30°W, with axis of relative rotation through top and bottom of projection. Details given in text.

America, together with existing data, provide an estimate of the mean Triassic pole for South America at 72°S, 91°W, N = 3, $\alpha_{95} = 15°$. The best-fit pole of relative rotation, obtained by fitting isobaths of the continental margins of South America and Africa (Bullard *et al.*, 1965), also superimposes the Triassic paleomagnetic poles of the two continents into a single grouping. This coincidence supports the assumption of a dipolar field in calculating paleomagnetic poles, and indicates that the geomagnetic field over South America and Africa was a dipolar field in Triassic time. Our results support the theories of continental drift and spreading of the sea floor. An alternative explanation for physical results of this type represents a major challenge to opponents of plate-tectonics theory.

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